

2.8 Division and modulo ^

Settings

Beta features

modulo operator / %

The modulo operator (%) evaluates the remainder of the division of two numbers. For example, $23 \% 10$ is 3.

Try 2.8.1: Example using expressions: Minutes to hours/minutes.

The program below reads the number of minutes entered by a user. The program then converts that number to hours and minutes.

Run the program, then modify the code to work in reverse: The user enters two numbers for hours and minutes, and the program outputs total minutes.



▶ Run

```
1  minutes = int(input("Enter minutes: "))
2  hours = minutes // 60
3  minutes_remaining = minutes % 60
4
5  print(minutes, "minute(s) is", hours, "hour(s) and", minutes_remaining, "minut
```

CONSOLE**CHALLENGE
ACTIVITY**

2.8.1: Enter the output of the integer expressions.

763650.1443318.qx3zqy7

Start

Type the program's output

```
x = 13
y = x / 2
print(y)
```

6.5

1

2

3

Check

Next

View solution ▼ (Instructors only)

Example 2.8.1: Getting digits.

Given a number, % and // can be used to get each digit. For a three-digit number user_val like 927

```
ones_digit    = user_val % 10    # Ex: 927 % 10 is 7.
tmp_val       = user_val // 10   #

tens_digit    = tmp_val % 10     # Ex: tmp_val = 927 // 10 is 92. Then 92 % 10
tmp_val       = tmp_val // 10

hundreds_digit = tmp_val % 10    # Ex: tmp_val = 92 // 10 = 9. Then 9 % 10 is
```

Example 2.8.2: Get prefix.

Given a 10-digit phone number stored as an integer, % and // can be used to get any part, such as 9365551212 (whose prefix is 555):

```
tmp_val = phone_num // 10000 # "// 10000" removes rightmost 4 digits , so "936
prefix_num = tmp_val % 1000 # "% 1000" gets rightmost 3 digits, so "936555 % 1000
```

Dividing by a power of 10 removes rightmost digits. Ex: 321 // 10 is 32. Ex: 321 // 100 is 3.

% by a power of 10 gets rightmost digits. Ex: 321 % 10 is 1. Ex: 321 % 100 is 21.

**CHALLENGE
ACTIVITY**

2.8.2: Division and modulo.

763650.1443318.qx3zqy7

Start

Integers `total_buttercups` and `num_buttercups` are read from input. Assign `num_bouquets` with bouquets that can be made with `total_buttercups` buttercups, if each bouquet consists of `num`.

[Click here for example](#) ▼

```
1 total_buttercups = int(input())
2 num_buttercups = int(input())
3
4 """ Your code goes here """
5
6 print("Maximum number of bouquets:", num_bouquets)
```

1

2

Check**Next level**[View solution](#) ▼ *(Instructors only)*